

Executive Summary — Surrogate Model Validation

Use case: UCFATIGUE_1 | 28 July 2026 | 2 model(s) evaluated

REQUIREMENTS PARTIALLY MET

Recommended model: GradientBoosting | Avg R²: 0.9643 | Q90: 5/7 | KS: 2/7

Project Overview		Reading the Table Below	
Use case	UCFATIGUE_1	R ²	Goodness of fit (higher = better)
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)	Q90	90th pct. relative error
Outputs	7: 1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro	Gap	Q90 – target (< 0 = margin)
		PASS	Q90 < 10%
		FAIL	Q90 ≥ 10%
Split	Train 80% / Val 10% / Test 10%		R ² ≥ 0.95
Requirement	Q90 < 10% per output		0.80 ≤ R ² < 0.95
			R ² < 0.80

Model Performance — GradientBoosting (Test Set)

Output	R ²	Q90	Target	Gap	Status
1g	0.9797	0.0257	10%	-0.0743	PASS
Vertical maneuver	0.9723	0.0318	10%	-0.0682	PASS
Vertical gust	0.9914	0.0262	10%	-0.0738	PASS
Turn	0.9286	0.0351	10%	-0.0649	PASS
Frontal gust	0.9964	0.2795	10%	+0.1795	FAIL
n0	0.9317	0.0149	10%	-0.0851	PASS
Giro	0.9499	0.2215	10%	+0.1215	FAIL

Analysis & Recommendations

- Significant failures (Frontal gust, Giro): Error is substantially above target. Investigate whether additional input features explain the residual variance, or consider a more expressive model architecture.
- Overfitting detected (1g, Vertical gust, Turn, n0, Giro): KS test rejects distributional equality between train and test residuals. Apply stronger regularisation (dropout, weight decay) or reduce model complexity.

Full technical details — scatter plots, KS overfitting test, split quality — in the [Technical Appendix](#) (page 3).

Technical Appendix

UCFATIGUE_1 — Full Validation Report

Technical Appendix — Contents

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Model Comparison (Test Set)

Model	Avg R ²	Q90 pass (7 outputs)	KS pass (7 outputs)
MLP	0.5237	5/7	7/7
GradientBoosting ★	0.9643	5/7	2/7

★ = recommended model. Q90 target: < 10% relative error. KS: $p \geq 0.05$ = no overfitting.

Full Metrics per Output

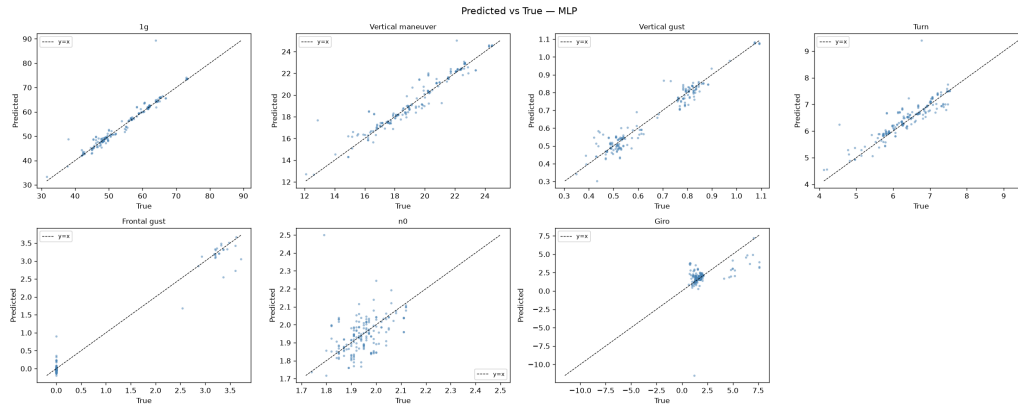
Output	MLP			GradientBoosting		
	R ²	Q90	Gap	R ²	Q90	Gap
1g	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682
Vertical gust	0.9377	0.0918	-0.0082	0.9914	0.0262	-0.0738
Turn	0.7544	0.0783	-0.0217	0.9286	0.0351	-0.0649
Frontal gust	0.9794	11.3300	+11.2300	0.9964	0.2795	+0.1795
n0	-0.7292	0.0723	-0.0277	0.9317	0.0149	-0.0851
Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215

pass Q90 < 10% fail / low R² R² ∈ [0.80, 0.95) Gap = Q90 − target (negative ⇒ margin).

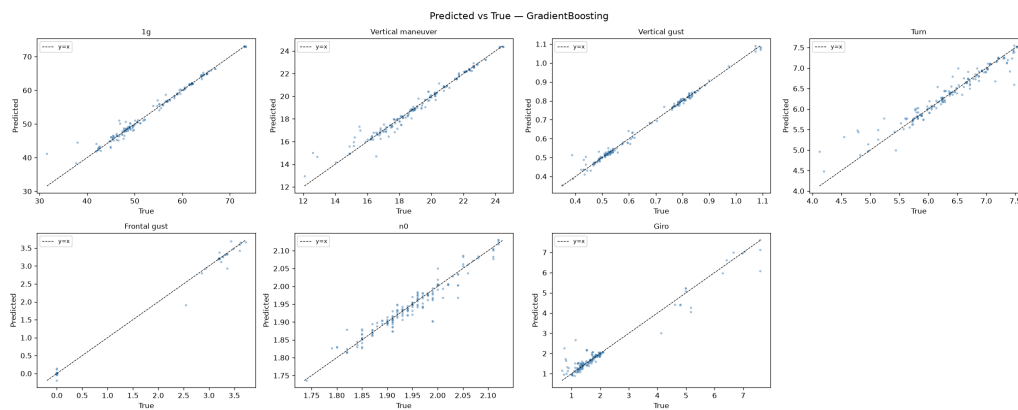
Predicted vs True (Scatter Plots, Test Set)

Each plot shows a sample of up to 5 000 test points. The dashed line represents perfect prediction ($\hat{y} = y$).

MLP



GradientBoosting



Overfitting Check — KS Test p-values

H_0 : residuals on train and test sets follow the same distribution. Rejection ($p < 0.05$) indicates overfitting.

Output	MLP	GradientBoosting
1g	0.473	0.004
Vertical maneuver	0.180	0.161
Vertical gust	0.635	0.042
Turn	0.078	0.044
Frontal gust	0.736	0.974
n0	0.378	0.032
Giro	0.781	0.017

Train / Test Split Quality

Metric	Value	Pass
Residual voxel proportion	0.000	✓
Valid test proportion	0.931	
Phacking test proportion	0.006	
Isolated test proportion	0.063	
Chi ² p-value	0.7969	✓

Residual voxel proportion ≤ 0.05 ensures test points cover the training space. Chi² $p \geq 0.05$ confirms the split is statistically unbiased.